# Reproducibility Report: Graham and Coppock (2021)

## Asking About Attitude Change

## Table of contents


**1 Paper Overview** **3**

**2 Summary** **3**

2.1 Does the deposited archive run? . . . . . 4

2.2 Does the maintained rewrite reproduce the paper? . . . . . 4

**3 Original Archive Reproducibility** **6**

**4 Ground Truth** **7**

4.1 Published claims, and the two instruments that check them . . . . . 9

**5 Maintained Rewrite** **10**

**6 Figures** **12**

**7 Rewrite Verification** **16**

**8 R Environment** **21**


*Drafted by Claude Opus 5 under the supervision of Alex Coppock.*

This repository holds the actively maintained replication code for Graham and Coppock (2021), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.


| | |
|---|---|
| Article | 10.1093/poq/nfab009 |
| Replication archive | 10.7910/DVN/GFF78K |

| | |
|---|---|
| Mirror deposit | 10.60600/YU/2ADPVK |
| Pre-analysis plans | none |
| Errata | `graham_coppock_2021_errata.pdf`, built from `errata.qmd` |

**The data are not redistributed here.** The deposit lives at Harvard Dataverse and that is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` records the Dataverse file identifiers, the UNF of each ingested data file, and two checksums per file: the MD5 of the bytes Dataverse serves, which is what this code was written against, and the MD5 Dataverse publishes. Here the two agree for all 12 files. They do not always, so the script verifies against the served bytes and reports any disagreement. It also checks byte sizes and stops if `original/` holds anything the manifest does not list, because an archive script that writes into its own directory adds files as well as overwriting them. Either way the exact bytes are pinned in version control even though the bytes themselves are not. Re-hosting a copy would create a second archive that can drift from the first, which is the problem this project exists to document rather than to add to.

**Repository layout.** `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it: `published_claims.csv` is the extraction of every numeric token in the article and its supplementary materials, and `build_ground_truth.R` assembles the comparison table by reading every rewrite value back out of `maintained/output/`, so it cannot drift from the scripts. `maintained/in_text_claims.R` is a second, independent instrument that recomputes each of those numbers by its own path and prints it beside the sentence that states it; the build compares the two and stops if they disagree. `errata.qmd` builds the note listing the sentences in the article that its own data do not support. `original/` is created by the download script and is deliberately absent from the repository. This README is the reproducibility report, also available as a PDF in `report/`.

**License.** CC0 1.0 Universal, matching the terms of the deposit this repository maintains, so nothing in the chain is more restrictive than the archive itself. See `LICENSE`.

**To reproduce.** Clone or download the repository, open `graham_coppock_2021.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposited archive from Dataverse, verifies its 12 checksums, produces every published figure, appendix table and in-text number into `maintained/output/`, rebuilds the ground truth from those outputs, runs the claims audit and its coverage gates, and re-checks the deposit. It takes about 75 seconds, of which roughly 65 are the 10,000-resample bootstrap

behind Table D.5. Individual scripts run on their own in any order, subject to the dependencies in each script's header: `clean_studies.R` first, then the two estimation scripts, then the figures, tables and in-text scripts that read them.

Required packages: tidyverse, estimatr, ggh4x, gridExtra, rsample, knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory and the scripts work equally well under `Rscript` outside RStudio. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check**, and the CSV and PNG output should come back byte-identical. The six PDF figures always show as changed, because a PDF records the time it was written; compare their PNG twins instead.

# 1 Paper Overview

**Citation**: Graham, M. H. and Coppock, A. (2021). Asking about attitude change. *Public Opinion Quarterly*, 85(1), 28–53.

**Research question**: When a survey asks people whether some event or piece of information made them more or less supportive of something, do their answers measure attitude change? The paper's claim is that they largely do not: respondents use the change question to express the *level* of their attitude, a bias the paper calls response substitution.

**Design**: Four survey experiments plus one illustrative survey. The illustration is a 4,034-respondent Lucid survey on impeaching Donald Trump, asked in the paper's proposed *counterfactual format*: respondents report their attitude, then report what they would have said had they not known about the Ukraine affair. Studies 1 (N = 417), 2a (N = 2,475) and 2b (N = 1,110) randomly assign respondents both to an information treatment and to a question format, so the self-reported change can be set against an experimental difference in means on the same treatment. Study 3 (N = 1,074) varies only the format. Estimation is by difference in means and by group means with HC2 standard errors, clustered by respondent where a respondent answers about more than one treatment.

**Main finding**: The change format performs badly and the counterfactual format performs well. Across ten treatments split by party, the more-minus-less statistic derived from the change format has the opposite sign to the experimental benchmark in 12 of 20 comparisons, against 3 of 20 for the counterfactual format. Asking a level question first cuts reports of any change by about 10 percentage points, which is what response substitution predicts.

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# 2 Summary

Two questions, answered before the detail.

## 2.1 Does the deposited archive run?

Not as deposited. `analysis_impeach.R`, which draws Figure 1, runs to completion. `analysis_main.R`, which produces everything else, fails twice.

- It fails first at `ggsave("drafts/POQ_RR/figures/substitution_new.png", ...)`, one of six active `ggsave()` calls writing into two directory trees on the authors' own machine that the deposit does not contain. Every analysis in the script completes; only the writes fail.
- With those directories created it fails again, and this one is substantive: `object 'check_labeller' not found`. The archive ships its own `facet_nested()` in `functions.R`, copied from an early standalone script, and that function calls `ggplot2:::check_labeller()` and `ggplot2:::var_list()`. Both are unexported internals and both have since been removed from ggplot2. The function is used for every panel of Figures 2, 4, 5 and E.2, so the failure takes out four of the paper's six figures.

Neither failure needed the authors' help or a file the archive omits. Substituting `ggh4x::facet_nested()`, the maintained successor to the same code, is enough to make the whole script run.

Two smaller things are worth recording because a reader reproducing this work will meet them. `analysis_impeach.R` writes `GrahamCoppock19-0280-Figure1.jpeg` to a bare relative path, and both scripts leave an `Rplots.pdf` behind when run non-interactively, so running the archive in its own directory silently adds two files to the deposit. And `functions.R` calls `plyr` in eight places without any script declaring it, while `analysis_main.R` loads `ggpubr`, `janitor` and `ggrepel` and never calls a function from any of them.

## 2.2 Does the maintained rewrite reproduce the paper?

Almost entirely. Every appendix table reproduces cell for cell. Of the 82 recorded claims, 67 compare a published number against a rewrite number and 62 of those match at the precision the page prints; 12 are claims about sign, shape or count rather than about a value, and 11 of those hold; the remaining 3 are magnitudes the article itself hedges, which are recorded without a verdict.

Table 1: Reproduction verdict by component

| Component | Verdict |
|---|---|
| Figure 1b (seven printed group means) | All seven reproduce |
| Figure 2 (ten printed labels) | All ten reproduce |
| Figure 3 (shape and the effects the text quotes) | Reproduces |
| Figure 4 (96 printed labels) | All 96 reproduce, one of them on a rounding tie |
| Figure 5 (24 printed labels) | All 24 reproduce |

| | |
|---|---|
| Figure E.2 (44 printed labels) | All 44 reproduce |
| Table D.3 (336 cells) | All reproduce, with the published column headings transposed |
| Table D.4 (216 cells, 72 p-values) | All reproduce |
| Table D.5 (60 point estimates) | All reproduce |
| Table D.5 (bootstrap intervals) | 53 of 60 rows; the deposited script reproduces all 60 |
| In-text numbers | All but two reproduce; both are the article rounding its own estimate |
| In-text claims about sign and shape | All but one hold; the exception is the sentence about the three sign misses |

The exceptions are small and none of them is a failure of the analysis.

**The article rounds two of its own numbers away from what its data give.** The text reports the two impeachment questions correlating at 0.82; the deposited data give 0.828, which prints as 0.83. It reports the level question reducing Democrats' reports of change by 14 points; the estimate is 13.5 points, which prints as 13. Both are one unit in the last printed digit, and neither touches a conclusion.

**Study 3's N is a count of answers, not of respondents.** The article reports 1,074. The deposited Study 3 extract carries 1,074 rows that answer the change question, which is the published figure, but only 1,023 distinct `id` values across 1,081 rows, so the `id` column is not a key and a count of distinct respondents gives 1,023 instead. The published Table D.3 settles which of the two the article means: its six Study 3 sample sizes sum to 1,074.

**Table D.3's column headings are transposed relative to its own body.** The published table heads its second and third value columns "No change" and "Less", but in 84 of its 84 rows the printed Difference equals More minus the column headed "No change", against 2 for the column headed "Less". Read in that corrected order all 336 cells reproduce. Read as printed, 172 do. The numbers in the table are right; the two headings are swapped.

**One printed label in Figure 4 sits on a rounding tie.** For the Undisputed accusation treatment among Republicans, the counterfactual-format estimate is a mean that comes to exactly -1.125. `lm_robust` returns it one unit in the last place away from that tie, so the deposited script prints it as -1.13. The published figure prints -1.12, which is what the tie itself gives, and which is also what the appendix's Table D.5 prints for the same quantity. The rewrite snaps each label's estimate and standard error to ten digits before printing it, which restores the tie, so it prints -1.12 as the figure does. Nothing about the estimate differs; only the last printed digit did.

**Table D.5's bootstrap intervals are the one place the rewrite and the deposit part company.** Both draw 10,000 resamples from seed 0. The deposited script resamples every cell in the counterfactual data and then keeps the twenty the table reports; the rewrite selects

the twenty first, so the two walk different random streams from the same seed. The deposited script reproduces all sixty published rows exactly. The rewrite differs on seven rows by a unit in the second decimal, which is Monte Carlo variation rather than a difference in the estimator, and no interval changes whether it covers zero.

**One sentence in the results section is wrong twice over.** The text says the counterfactual format gets the sign wrong in 3 of 20 cells, and that in all three neither the counterfactual estimate nor the difference in means can be distinguished from zero. The count is 2, not 3: one of the three cells has a counterfactual estimate of exactly zero, and a zero has no sign to be opposite. Comparing the two signs directly, which is what the published count does, returns 3 under the linear algebra the deposited archive was run against and 4 under a current one, so that figure is not stably reproducible in either direction. Of the two cells that survive, one is null and one is not: the counterfactual estimate for the disputed accusation among Democrats has a 95 per cent interval that excludes zero in Figure 4, in Table D.5, and in the rewrite alike. Those two corrections to the one sentence, the two rounded numbers above, and the transposed Table D.3 headings are the 5 claims whose fault lies with the article rather than with any code, and they are collected in `graham_coppock_2021_errata.pdf` at the root of this repository, which quotes each published sentence and prints the corrected one with every number in it computed at render time. The note also corrects 10 errors in the reference list, found by checking every entry against Crossref and against the cited works themselves.

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## 3 Original Archive Reproducibility

**Archive source**: Harvard Dataverse, DOI 10.7910/DVN/GFF78K, 12 files.

The archive contains two analysis scripts, a shared `functions.R`, a codebook, a readme, and seven data files: the merged Studies 1 to 3 frame and the impeachment survey as `.rds`, and both plus the four individual studies as `.csv`.

Table 2: Original archive reproducibility check (current R environment)

| Script | Status | Issue |
|---|---|---|
| functions.R | Sourced by both | Custom facet_nested() calls two ggplot2 internals that no longer exist; undeclared dependency on plyr |
| analysis_impeach.R | Runs | Writes a JPEG to a bare relative path, and an Rplots.pdf under Rscript |
| analysis_main.R | FAILS | Six ggsave() calls to directories the deposit does not contain, then facet_nested() |

**Failure details.**

- `analysis_main.R` lines 192, 193, 531, 545, 551 and 566 call `ggsave()` into `drafts/POQ_RR/figures/` and `drafts/POQ_final/final figures/`. Neither directory is in the deposit. The script's own comments elsewhere say to change the paths and uncomment the calls, and most of the `ggsave()` calls are indeed commented out; these six are not.
- `functions.R` defines a 250-line `facet_nested()` that predates the version now in `ggh4x`. It calls `ggplot2:::check_labeller()` at line 159 and `ggplot2:::var_list()` at line 335. Both were unexported and both are gone from current ggplot2, and either is fatal. `ggh4x::facet_nested()` is the maintained successor and is a drop-in replacement here.
- `functions.R` also defines a `position_jitter_ellipse()` ggproto that calls `ggplot2:::with_seed_null()`. That internal still exists, so Figure 1 draws today. It is an unexported function nonetheless, and the rewrite uses the standard `position_jitter()` instead; the panel carries no numbers and the article describes the jitter only as "a small amount random noise added to distinguish the points."

**Deprecations that warn rather than fail**: `%>%`, `do(tidy(...))`, `gather()` and `spread()`, `geom_errorbar()` for confidence bars, `size =` for line widths, `ifelse()`, `rm(list = ls())` at the top of both scripts, and `source("functions.R")` by relative path.

**One coding slip, carried forward rather than corrected.** Assembling the estimates for Figures 2, 4, 5 and E.2, the deposited script sorts the topics with `factor(Study, c(1, 4, 2, 3))`. The study labels are `1`, `2a`, `2b` and `3`, so `2a` and `2b` do not match any level and become `NA`, sorting last. That ordering is what produced the published facet sequence, so the rewrite keeps it; it uses the sort key and then drops it, rather than carrying a study column that is empty for half the rows.

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## 4 Ground Truth

`ground_truth/build_ground_truth.R` writes `graham_coppock_2021_ground_truth.csv` and runs as the last step of `run_all.R`, so the table cannot drift from the pipeline. Published values are read from the article and the supplementary materials and live either in that script or in the four `published_*.csv` files beside it, which transcribe Tables D.3, D.4 and D.5 and the printed labels of Figures 2, 4, 5 and E.2 in the layout the pages print them in. Every value in the `value_rewrite` column is read back out of `maintained/output/`; none is typed. The verdicts are computed, by printing the rewrite's number to the precision the page prints and comparing digits.

`ground_truth/extract_archive_values.R` fills the `value_script` column. It runs the deposited scripts in a scratch copy, with the two repairs above, and compares each of their objects against the maintained output it corresponds to.

Table 3: Deposited scripts against the maintained rewrite, object by object

| Object | Archive rows | Rewrite rows | Joined | Largest difference |
|---|---|---|---|---|
| grand comparison estimates | 480 | 480 | 480 | 0.0000 |
| pretreated estimates | 428 | 428 | 428 | 0.0000 |
| Figure 1b group means | 7 | 7 | 7 | 0.0000 |
| Figure 3a estimates | 30 | 40 | 30 | 0.0000 |
| Table D.3 cells | 336 | 336 | 336 | 0.0000 |
| Table D.4 cells | 216 | 216 | 216 | 0.0000 |
| Table D.5 point estimates | 60 | 60 | 60 | 0.0000 |
| Table D.5 bootstrap quantities | 60 | 60 | 60 | 0.0109 |

Everything except the Table D.5 bootstrap agrees to machine precision. The Figure 3a row joins 30 of the rewrite's 40 estimates because the rewrite also estimates the pure independents that the panel leaves out and the surrounding text quotes.

Table 4: Ground truth: 76 of 82 rows record no disagreement between the article and the rewrite

| Where the difference lies | Rows |
|---|---|
| no disagreement | 76 |
| paper_internal | 5 |
| rewrite | 1 |

Table 5: The 6 claims the rewrite does not reproduce, and where the fault lies

| Claim | Published | Rewrite | Locus |
|---|---|---|---|
| Text, p. 32: Correlation of the two impeachment questions | 0.82 | 0.8285 | paper_internal |
| Text, p. 44: Same decrease among Democrats (points) | 14 | 13.4590 | paper_internal |
| Text, p. 46: Counterfactual format has the opposite sign | 3 | 2.0000 | paper_internal |
| Text, p. 46: Sign-miss cells in which neither estimate is distinguishable from zero | 3 | 1.0000 | paper_internal |

| | | | |
|---|---|---|---|
| Table D.3: Cells agreeing if the column headings are read as printed | 336 | 172.0000 | paper_internal |
| Table D.5: Bootstrap standard errors and interval endpoints agreeing | 60 | 53.0000 | rewrite |

Every row that fails carries a locus, because a bare zero reads as a failure of the rewrite and here it is one only once, on the Table D.5 bootstrap intervals. The build stops on any row carrying a verdict without a locus or a locus without a verdict.

The full table, all 82 rows with their notes, is `ground_truth/graham_coppock_2021_ground_truth.csv`.

## 4.1 Published claims, and the two instruments that check them

The table above starts from the pipeline and asks what each output corresponds to in the article. That question cannot find a claim the pipeline never reaches, so the repository also works in the other direction. `ground_truth/published_claims.csv` is an exhaustive extraction: every numeric token in the body, the footnotes, the tables, the figures and the supplementary materials, read off the typeset page rather than searched for by pattern, together with a companion pass for numbers the article spells out in words. Each of the 177 resulting claims is classified by hand.

Table 6: The extraction: 177 published claims by type

| Claim type | Claims | Needs a block |
|---|---|---|
| pipeline | 64 | 64 |
| structural | 41 | 16 |
| definitional | 32 | 9 |
| transcribed | 25 | 0 |
| descriptive | 15 | 15 |

A `pipeline` claim is one the analysis code produces; a `descriptive` claim is about sign, shape or count rather than about a value. Each of those 79 needs a row in the ground truth. A `definitional` or `structural` claim (a scale endpoint, a panel count, a field date) gets a block only where the pipeline can reach the quantity, and a `transcribed` claim is quoted from another source and cannot drift. All told 104 of the 177 claims require a block.

`maintained/in_text_claims.R` carries one block per such claim. Each block quotes the article's sentence verbatim, then recomputes the number from `maintained/output/`, including the cleaned data written there, and prints it at the precision the page uses, with the evidence beneath it where the claim is about shape rather than value. It never reads the ground truth, so the two instruments reach each number by separate paths, and `build_ground_truth.R`

runs it as a program, counts what it printed, and compares the two value by value. The build halts if the claims script prints a claim the extraction does not list, if the extraction lists a claim the script does not print, if a `pipeline` or `descriptive` claim has no row in the ground truth, or if the two instruments disagree on any number. All 104 claims that require a block print one.

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# 5 Maintained Rewrite

Table 7: Maintained rewrite scripts

| Script | Output |
|---|---|
| helpers.R | Packages, colours, theme, the shared grand comparison panel |
| clean_studies.R | output/data_study1-3_clean.rds, output/data_impeach_clean.rds |
| estimates_grand_comparison.R | output/estimates_grand_comparison.{rds,csv} |
| estimates_pretreated.R | output/estimates_pretreated.{rds,csv} |
| figure_1_impeach_counterfactual.R | output/figure_1_impeach_counterfactual.{pdf,png,csv} |
| figure_2_cornish_example.R | output/figure_2_cornish_example.{pdf,png,csv} |
| figure_3_response_substitution.R | output/figure_3_response_substitution.{pdf,png}, three CSVs |
| figure_4_grand_comparison.R | output/figure_4_grand_comparison.{pdf,png,csv} |
| figure_5_pretreat_comparison.R | output/figure_5_pretreat_comparison.{pdf,png,csv} |
| figure_e2_study2b_simultaneous.R | output/figure_e2_study2b_simultaneous.{pdf,png,csv} |
| table_d3_change_distribution.R | output/table_d3_change_distribution{,_estimates}.csv |
| table_d4_counterfactual_accuracy.R | output/table_d4_counterfactual_accuracy{,_estimates}.csv |
| table_d5_ate_vs_self.R | output/table_d5_ate_vs_self{,_estimates}.csv |
| text_intext_stats.R | output/text_intext_stats.csv |
| text_sign_comparison.R | output/text_sign_comparison.csv |

| | |
|---|---|
| in_text_claims.R | Printed to the console: every published number beside its sentence |

Two structural choices are worth naming. Figures 2, 4, 5 and E.2 are the same panel drawn over different subsets, which is how the deposit built them too, so the panel is one function in `helpers.R` and every figure script calls it. And the estimates behind those four figures are computed once, in `estimates_grand_comparison.R` and `estimates_pretreated.R`, so no estimate in the paper is computed twice by two scripts that could then disagree.

Every figure script writes a CSV of what it plots, and every table script writes its cells twice: once formatted as the appendix lays them out, once unrounded. The unrounded copy is what the ground truth reads, so a comparison against a published two-decimal table never has to parse a three-decimal string and round it a second time.

Table 8: Deprecated patterns and replacements

| Original | Rewrite |
|---|---|
| custom facet_nested() using ggplot2 internals | ggh4x::facet_nested() |
| coord_flip() with Format on the discrete axis | native horizontal layout: y = Format, x = estimate |
| position_jitter_ellipse() ggproto | position_jitter() with a seed |
| %>% | |> |
| do(tidy(...)) | reframe(tidy(...)) |
| gather() / spread() | pivot_longer() / pivot_wider() |
| geom_errorbar(width = 0) | geom_linerange() |
| ifelse() | if_else() |
| recode() | recode_values() |
| replicate(10000, oneDifference(.)) | rsample::bootstraps(times = 10000) |
| xtable + print.xtable to the console | write_csv() to output/ |
| size = for line widths | linewidth = |
| rm(list = ls()) | removed |
| ggsave() to the authors' own directories | ggsave() through here::here() into maintained/output/ |
| library() in each script | source(helpers.R); all packages in helpers.R |

**The layout change is the largest one.** The deposit draws the grand comparison panel with `coord_flip()` over `facet_nested(Party + Estimator ~ Topic, switch = "y")`. Combined with `space = "free_y"`, that arrangement errors out under `ggh4x`. The rewrite maps the panel natively instead, with `Format` on the vertical axis and the estimate on the

horizontal, and facets on `Topic ~ Party + Estimator` with `switch = "x"`. Nothing about the estimates changes and the figures carry the same content in the same arrangement.

**The bootstrap is the same estimator by a different engine.** The deposit resamples rows by hand inside `replicate()`; the rewrite uses `rsample::bootstraps()` at the same seed and the same 10,000 draws. Both are the nonparametric bootstrap over the same rows, and the point estimates are identical. The intervals differ on seven of sixty rows in the second decimal, for the reason given in the Summary.

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## 6 Figures

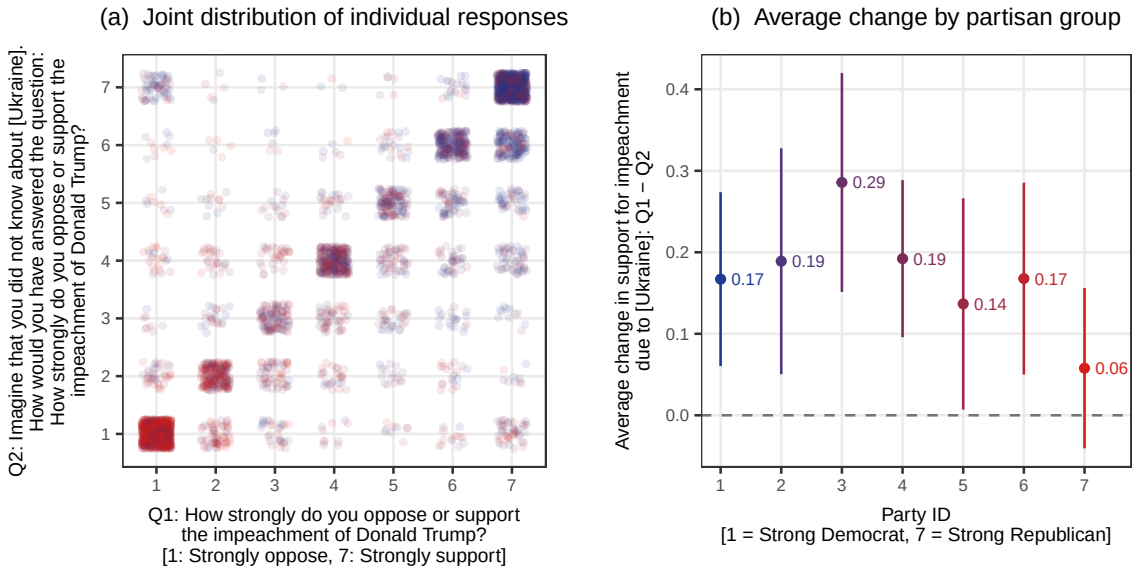



Figure 1: Figure 1. Counterfactual format example: the joint distribution of the two impeachment questions, and the average change by partisan group.

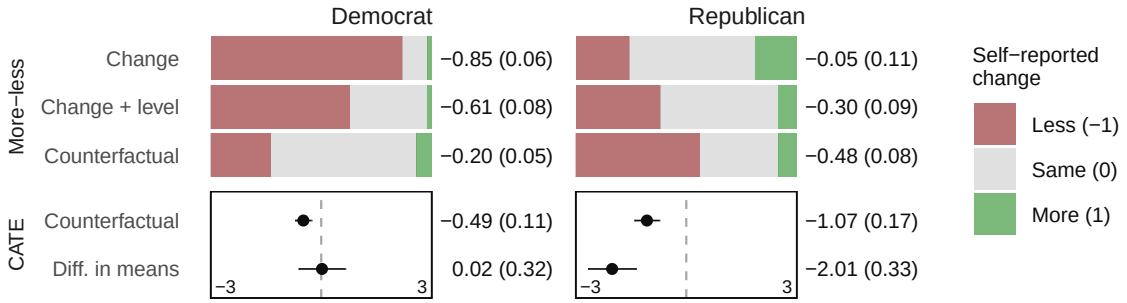



Figure 2: Figure 2. Self-reported change in the Cornish example, by party: three question formats against the experimental difference in means.

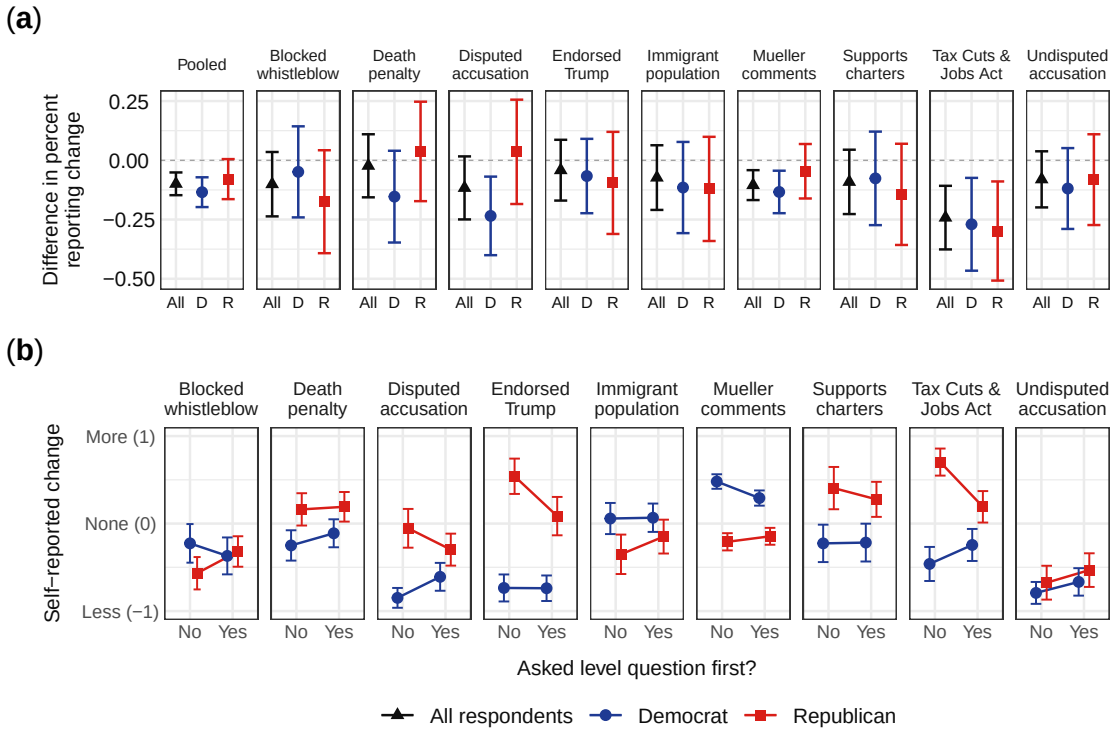



Figure 3: Figure 3. The effect of asking the level question first, on reporting any attitude change (a) and on the sign of self-reported change (b).

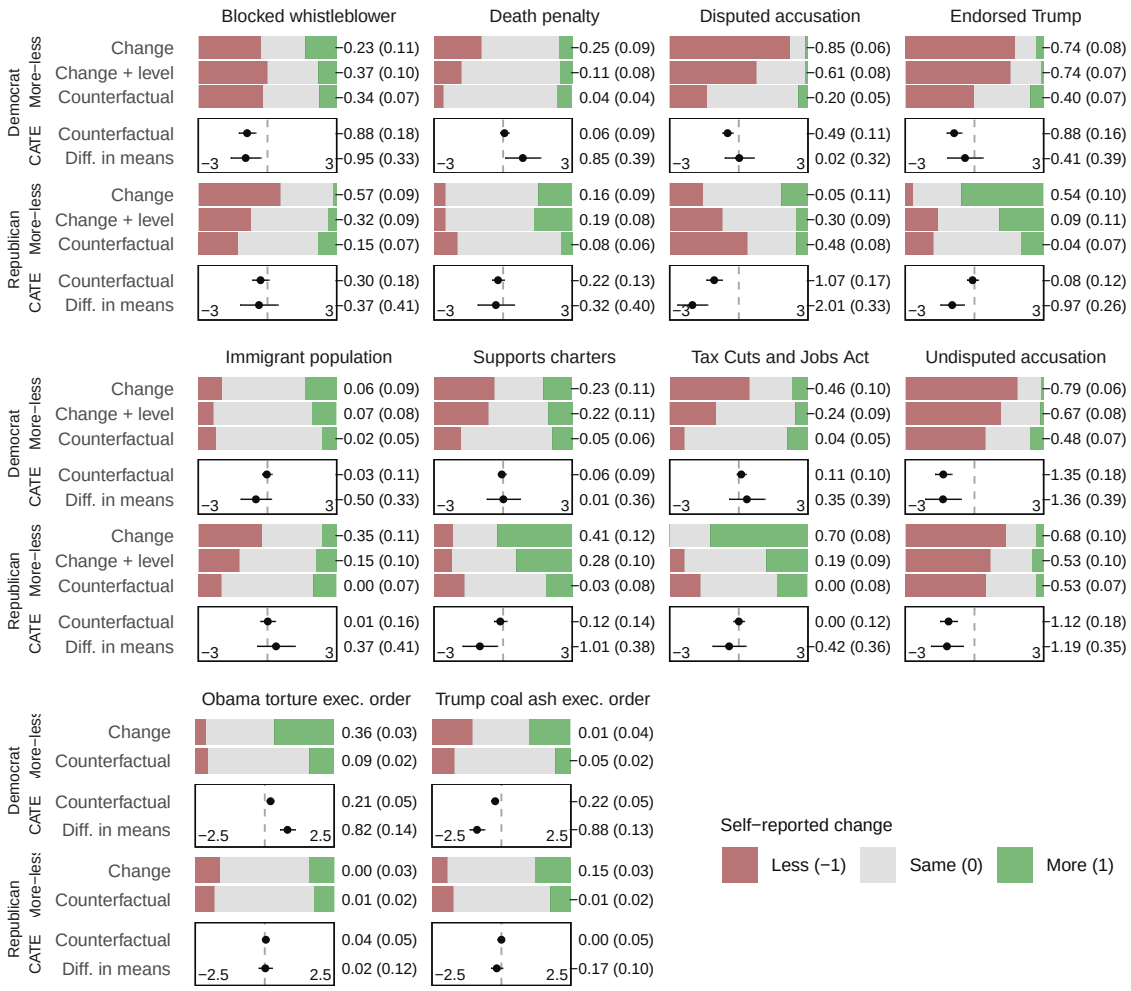



Figure 4: Figure 4. The full comparison of the change format, the randomized counterfactual format and the experiment, over the ten treatments in Studies 1 and 2a.

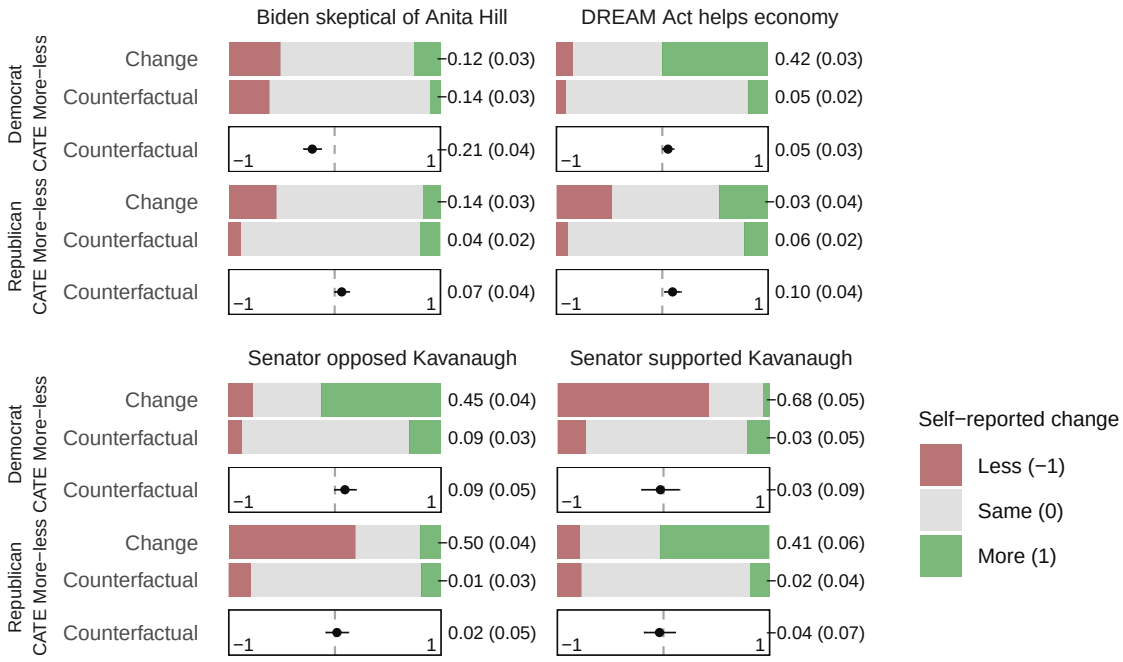



Figure 5: Figure 5. The change format against the nonrandomized counterfactual format, for the Study 2a treatments subjects may have met before the survey.

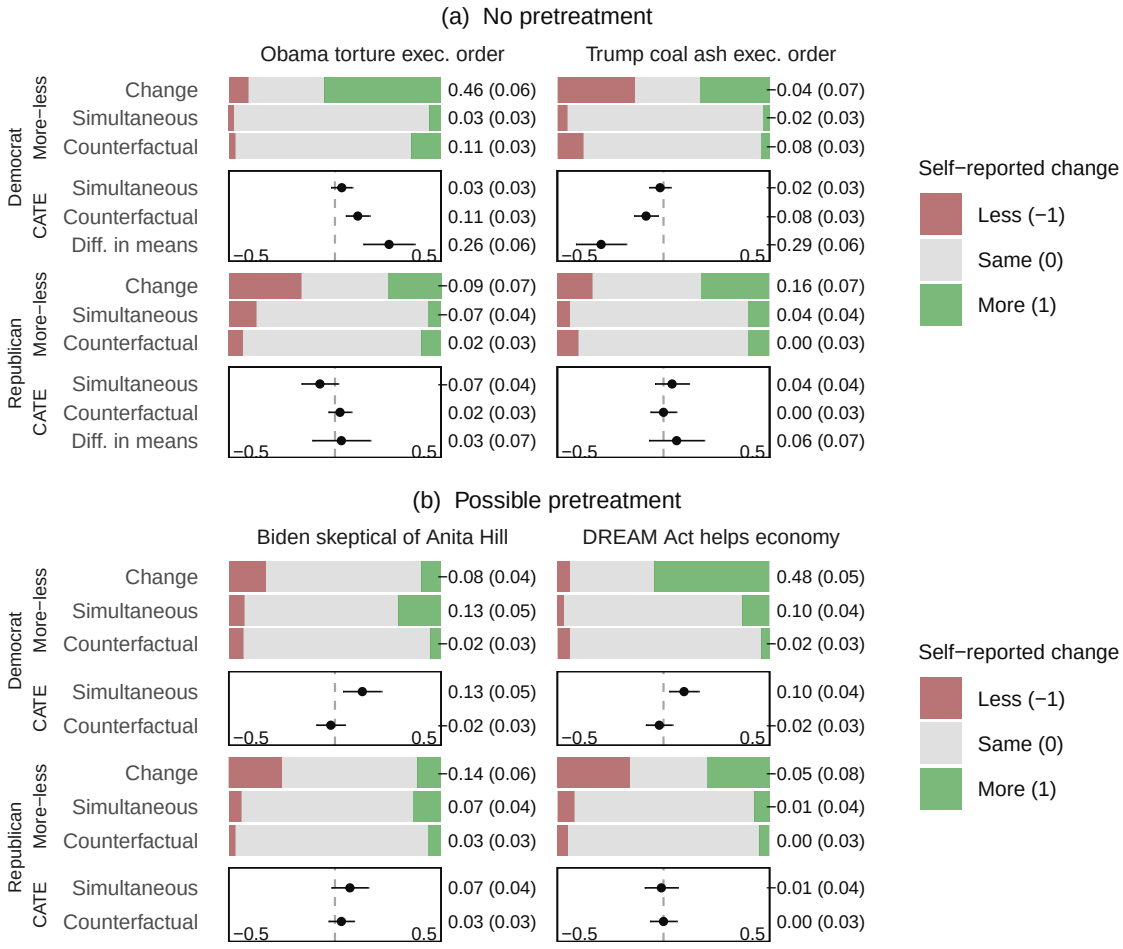



Figure 6: Figure E.2. Study 2b, setting the simultaneous outcomes format against the change and counterfactual formats.

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# 7 Rewrite Verification

Table 9: Rewrite verification: 62 of 67 value claims match the published values, and 11 of 12 descriptive claims hold

| Table or figure | Claim | Published | Rewrite | Match | Holds | Locus |
|---|---|---|---|---|---|---|
| Text, p. 31 | Survey responses obtained from Lucid | 4034 | 4034.0000 | 1 | NA | NA |

| | | | | | | |
|---|---|---|---|---|---|---|
| Text, p. 32 | Correlation of the two impeachment questions | 0.82 | 0.8285 | 0 | NA | paper_inte |
| Text, p. 32 | Share reporting exactly zero change (per cent) | 72 | 72.0870 | 1 | NA | NA |
| Text, p. 32 | Average difference between the two questions | 0.16 | 0.1601 | 1 | NA | NA |
| Text, p. 32 | Standard error of that average | 0.02 | 0.0226 | 1 | NA | NA |
| Text, p. 38 | Cornish, Democrats reporting less likely (per cent) | 87 | 86.7920 | 1 | NA | NA |
| Text, p. 38 | Cornish, Republicans reporting no effect (per cent) | 57 | 56.7570 | 1 | NA | NA |
| Text, p. 39 | Cornish, Democrats' more-minus-less score, change format | -0.85 | -0.8491 | 1 | NA | NA |
| Text, p. 39 | The same score with the level question asked first | -0.61 | -0.6087 | 1 | NA | NA |
| Text, p. 39 | Cornish, effect of the level question on Democrats' more-minus-less score | 0.24 | 0.2404 | 1 | NA | NA |
| Text, p. 39 | Standard error of the level question's effect, Democrats | 0.10 | 0.0972 | 1 | NA | NA |
| Text, p. 39 | Cornish, same effect among Republicans | -0.24 | -0.2438 | 1 | NA | NA |
| Text, p. 39 | Standard error of the level question's effect, Republicans | 0.14 | 0.1421 | 1 | NA | NA |
| Text, p. 39 | Cornish, difference in means among Democrats | 0.02 | 0.0208 | 1 | NA | NA |
| Text, p. 39 | Standard error of that difference in means, Democrats | 0.32 | 0.3186 | 1 | NA | NA |
| Text, p. 39 | Cornish, difference in means among Republicans | -2.01 | -2.0131 | 1 | NA | NA |
| Text, p. 39 | Standard error of that difference in means, Republicans | 0.33 | 0.3263 | 1 | NA | NA |
| Text, p. 39 | Parties of two whose experimental estimate is negative with an interval excluding zero | 1 | 1.0000 | NA | TRUE | NA |
| Text, p. 39 | Cornish, counterfactual format estimate among Democrats | -0.49 | -0.4854 | 1 | NA | NA |
| Text, p. 39 | Standard error of the counterfactual estimate, Democrats | 0.11 | 0.1111 | 1 | NA | NA |

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|---|---|---|---|---|---|---|
| Text, p. 40 | Cornish, counterfactual format estimate among Republicans | -1.07 | -1.0685 | 1 | NA | NA |
| Text, p. 40 | Standard error of the counterfactual estimate, Republicans | 0.17 | 0.1693 | 1 | NA | NA |
| Text, p. 40 | Democrats overstate negative change by approximately half a point | 0.5 | 0.5062 | NA | NA | NA |
| Text, p. 40 | Republicans understate negative change by almost a full point | 1 | 0.9446 | NA | NA | NA |
| Text, p. 40 | Study 1 N | 417 | 417.0000 | 1 | NA | NA |
| Text, p. 40 | Study 2a N | 2475 | 2475.0000 | 1 | NA | NA |
| Text, p. 44 | Study 2b N | 1110 | 1110.0000 | 1 | NA | NA |
| Text, p. 44 | Study 3 N | 1074 | 1074.0000 | 1 | NA | NA |
| Text, p. 44 | Decrease in reporting any change, all respondents (points) | 10 | 9.9271 | 1 | NA | NA |
| Text, p. 44 | Standard error of that effect (points) | 2 | 2.4431 | 1 | NA | NA |
| Text, p. 44 | Same decrease among Democrats (points) | 14 | 13.4590 | 0 | NA | paper_inte |
| Text, p. 44 | Standard error, Democrats (points) | 3 | 3.1890 | 1 | NA | NA |
| Text, p. 44 | Same decrease among Republicans (points) | 8 | 7.9455 | 1 | NA | NA |
| Text, p. 44 | Standard error, Republicans (points) | 4 | 4.2791 | 1 | NA | NA |
| Text, p. 44 | Same decrease among pure independents (points) | 2 | 2.2329 | 1 | NA | NA |
| Text, p. 44 | Standard error, independents (points) | 6 | 6.1777 | 1 | NA | NA |
| Text, p. 44 | Party groups of two whose decrease exceeds the independents' | 2 | 2.0000 | NA | TRUE | NA |
| Text, p. 46 | Cells compared, change format against the experiment | 20 | 20.0000 | 1 | NA | NA |
| Text, p. 46 | Change format has the opposite sign | 12 | 12.0000 | 1 | NA | NA |
| Text, p. 46 | Counterfactual format has the opposite sign | 3 | 2.0000 | 0 | NA | paper_inte |
| Text, p. 46 | Sign-miss cells in which neither estimate is distinguishable from zero | 3 | 1.0000 | NA | FALSE | paper_inte |
| Text, p. 46 | Tax Cuts and Jobs Act parties of two whose experimental estimate contains zero | 2 | 2.0000 | NA | TRUE | NA |

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|---|---|---|---|---|---|---|
| Text, p. 46 | Opportunities to evaluate a counterfactual guess | 40 | 40.0000 | 1 | NA | NA |
| Text, p. 47 | Difference in means tests rejecting at $p < 0.05$ | 12 | 12.0000 | 1 | NA | NA |
| Text, p. 47 | The same count as a percentage of the forty tests | 30.0 | 30.0000 | 1 | NA | NA |
| Text, p. 46 | Comparisons of the experiment against the counterfactual guess | 20 | 20.0000 | 1 | NA | NA |
| Text, p. 46 | Differences whose bootstrap interval excludes zero | 6 | 6.0000 | 1 | NA | NA |
| Text, p. 46 | The same count as a percentage of the twenty comparisons | 30 | 30.0000 | 1 | NA | NA |
| Text, p. 49 | Kavanaugh cells of four in which most respondents report the vote changed their preference | 4 | 4.0000 | NA | TRUE | NA |
| Text, p. 49 | Sign of the counterfactual estimate for Democrats whose Senator opposed Kavanaugh | 1 | 1.0000 | NA | TRUE | NA |
| Text, p. 49 | Parties of two whose counterfactual estimate of Biden's loss excludes zero | 1 | 1.0000 | NA | TRUE | NA |
| Text, p. 49 | Of one: whether the DREAM Act counterfactual boost is larger for Republicans | 1 | 1.0000 | NA | TRUE | NA |
| Figure 1b | Average change, party ID 1 | 0.17 | 0.1670 | 1 | NA | NA |
| Figure 1b | Average change, party ID 2 | 0.19 | 0.1890 | 1 | NA | NA |
| Figure 1b | Average change, party ID 3 | 0.29 | 0.2857 | 1 | NA | NA |
| Figure 1b | Average change, party ID 4 | 0.19 | 0.1921 | 1 | NA | NA |
| Figure 1b | Average change, party ID 5 | 0.14 | 0.1367 | 1 | NA | NA |
| Figure 1b | Average change, party ID 6 | 0.17 | 0.1677 | 1 | NA | NA |
| Figure 1b | Average change, party ID 7 | 0.06 | 0.0578 | 1 | NA | NA |
| Figure 1b | Partisan groups of seven whose average change is positive with an interval excluding zero | 6 | 6.0000 | NA | TRUE | NA |
| Figure 3a | Estimates plotted (ten facets by All, Democrat, Republican) | 30 | 30.0000 | 1 | NA | NA |
| Figure 3a | Pooled decrease in reporting any change, all respondents (points) | 10 | 9.9271 | 1 | NA | NA |
| Figure 3b | Means plotted (nine facets by two formats by two parties) | 36 | 36.0000 | 1 | NA | NA |

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|---|---|---|---|---|---|---|
| Figure 3b | Endorsed Trump, Republicans: sign of the level question's effect | -1 | -1.0000 | NA | TRUE | NA |
| Figure 3b | Mueller comments, Democrats: sign of the level question's effect | -1 | -1.0000 | NA | TRUE | NA |
| Figure 2 | Printed labels agreeing with the published figure | 10 | 10.0000 | 1 | NA | NA |
| Figure 4 | Printed labels agreeing with the published figure | 96 | 96.0000 | 1 | NA | NA |
| Figure 5 | Printed labels agreeing with the published figure | 24 | 24.0000 | 1 | NA | NA |
| Figure E.2a | Printed labels agreeing with the published figure | 24 | 24.0000 | 1 | NA | NA |
| Figure E.2b | Printed labels agreeing with the published figure | 20 | 20.0000 | 1 | NA | NA |
| Figure 4 | Undisputed accusation, Republican, CATE, Counterfactual | -1.12 | -1.1250 | 1 | NA | NA |
| Table D.3 | Cells agreeing with the published table | 336 | 336.0000 | 1 | NA | NA |
| Table D.3 | Cells agreeing if the column headings are read as printed | 336 | 172.0000 | 0 | NA | paper_inte |
| Table D.3 | Sample sizes agreeing | 84 | 84.0000 | 1 | NA | NA |
| Table D.4 | Cells agreeing with the published table | 216 | 216.0000 | 1 | NA | NA |
| Table D.4 | p-values agreeing | 72 | 72.0000 | 1 | NA | NA |
| Table D.4 | Sample sizes agreeing | 72 | 72.0000 | 1 | NA | NA |
| Table D.5 | Point estimates agreeing with the published table | 60 | 60.0000 | 1 | NA | NA |
| Table D.5 | Sample sizes agreeing | 60 | 60.0000 | 1 | NA | NA |
| Table D.5 | Bootstrap standard errors and interval endpoints agreeing | 60 | 53.0000 | 0 | NA | rewrite |
| Figure E.2a | No-pretreatment cells of four in which the simultaneous format sits further from the experiment | 3 | 3.0000 | NA | TRUE | NA |
| Figure E.2b | Ratio of the simultaneous to the change format share of Democrats reporting more support | 2 | 2.2177 | NA | NA | NA |

# 8 R Environment

Table 10: Key package versions

| Package | Version |
|---|---|
| tidyverse | 2.0.0 |
| estimatr | 2.0.1 |
| ggh4x | 0.3.1 |
| gridExtra | 2.3 |
| rsample | 1.3.2 |
| knitr | 1.51 |
| kableExtra | 1.4.0 |
| here | 1.0.2 |

R version: R version 4.6.0 (2026-04-24)